

Assignment: Object-Oriented Programming in Python

## Problem Statement

### Question 1: Student Class

Create a class `Student` that:

Has attributes: `name`, `roll_no`

Has a method `display()` to print student details  
Create one object and call the method.

### Question 2: Simple Calculator

Create a class `Calculator` that:

Has two numbers

Has methods: `add()` and `subtract()`  
Create an object and perform both operations.

### Question 3: Bank Account

Create a class `BankAccount` that:

Has `account_no` and `balance`

Has methods `deposit()` and `withdraw()`  
Perform one deposit and one withdrawal.

### Question 4: Rectangle

Create a class `Rectangle` that:

Has `length` and `breadth`

Has a method `area()` to calculate area

### Question 5: Constructor Practice

Create a class `Car` that:

Uses `__init__()` to set `brand` and `price`

Prints car details using a method

## Question 6: Inheritance (Very Simple)

Create a class `Animal` with a method `sound()`

Create a class `Dog` that inherits `Animal`

Call the method using a `Dog` object

## Question 7: Polymorphism (Easy)

Create two classes `Cat` and `Cow`

Both should have a method `sound()`

Call both methods using objects

## Question 8: `__str__()` Method

Create a class `Book` that:

Has `title` and `author`

Uses `__str__()` to print book details

## Question 9: Object Counter

Create a class that:

Counts how many objects are created

Print the total count

## Question 10: Mini Task

Create a class `Mobile` that:

Has `brand` and `price`

Has a method `show()`

Create two objects and display details.